



翔名科技股份有限公司
Feedback Technology Corp.

Automatic Pre-treatment & Line Expansion for ENP Process

Background

- **Customer:** AMAT.
- **Current Weakness:** Traditional manual chemical ENP faces bottlenecks in efficiency, stability, and cost control.
- **Requirement:** Increase ENP capacity.
- **Benefits:** Upgrading to automation drives significant improvements in productivity, quality, safety and cost-efficiency.

- 1

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ADDITIONAL SINGLE-ARM CRANE SIDE VIEW

The top plan view shows the layout of the crane with the following dimensions and labels:

- Overall length: 18006.09
- Segment lengths: 2498.09, 3003, 2500, 3703, 2400, 3902
- Labels from left to right: WATER RINSE TANK, WATER RINSE TANK, WATER RINSE TANK, WATER RINSE TANK, WATER RINSE TANK, ACID PICKLING TANK, ACID PICKLING TANK, ULTRASONIC WATER TANK, WATER RINSE TANK, ACTIVATION TANK, ACTIVATION TANK, DEGREASING TANK, SLOPED WALKWAY, LOADING AREA
- Bottom labels (1, 5, 4, 3, 2, 1) correspond to the following components:
 - 1: WATER SPRAY SOLENOIDSOLENOID VALVE
 - 5: WATER SPRAY SOLENOIDSOLENOID VALVE
 - 4: WATER SPRAY SOLENOIDSOLENOID VALVE
 - 3: WATER SPRAY SOLENOIDSOLENOID VALVE
 - 2: WATER SPRAY SOLENOIDSOLENOID VALVE
 - 1: WATER SPRAY SOLENOIDSOLENOID VALVE
 - Water Spray Solenoid Valve
 - Water Inlet Valve
 - Water Inlet Valve
 - Water Inlet Valve
 - Water Spray Solenoid Valve
- Dimension 13000 is indicated for the bottom section.

The two side elevation views show the crane's profile with the following dimensions:

- Left Elevation:**
 - Overall height: 1086
 - Top width: 1173
 - Inner width: 681
 - Bottom width: 780
 - Height from bottom to top of tank: 900
 - Height from bottom to top of tank: 900
- Right Elevation:**
 - Overall height: 900
 - Top width: 1234
 - Inner width: 1074
 - Bottom width: 460
 - Height from bottom to top of tank: 900

Benefits

Current Status (Manual Operations)

Future State (Automated Operations)

Industrial Safety Risk



Critical Risk. Direct contact with hazardous chemical. Splashing and inhalation danger.



Risk Avoidance. Zero personnel contact. Complete hazard isolation.

Manpower Deployment

Exponential demand with scale. Major recruitment challenges.

Lower demand. Full system takeover of all operations.

Quality Stability

Experience Dependent. High batch consistency fluctuations.

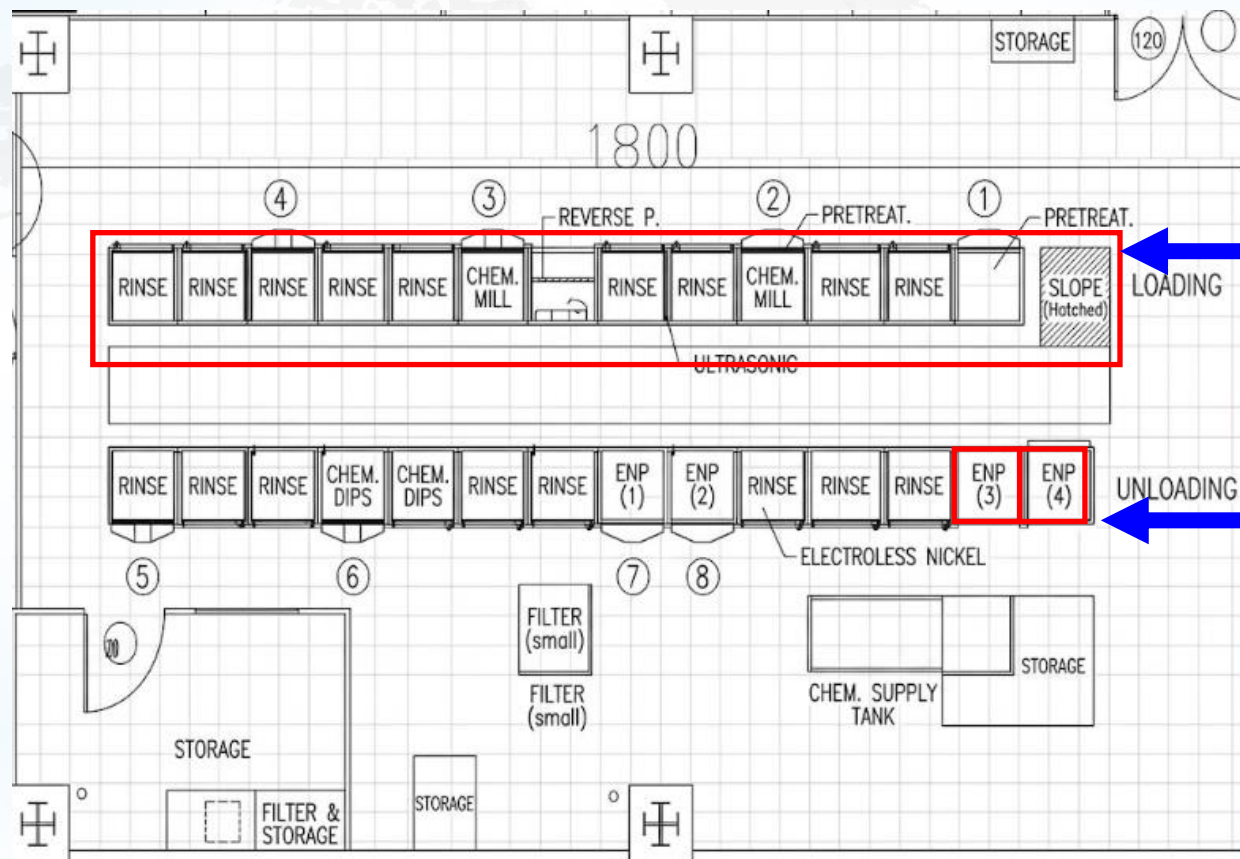
Process stability. Automated parameters ensure high quality yield.

Cost Control

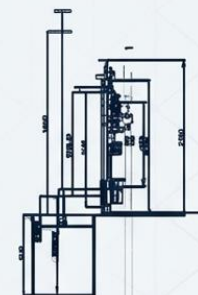
High Hidden Costs. Protective gear, high turnover, re-work.

Optimal Costs. Lower unit cost after Capex, optimized profitability.

Planned Layout Changes

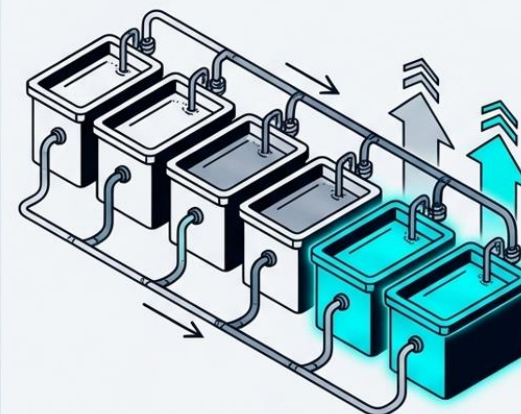


SINGLE-ARM JIB CRANE - SIDE VIEW



SYSTEM OPTIMIZATION: NEW PLATING CAPACITY

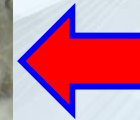
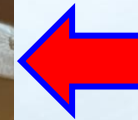
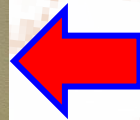
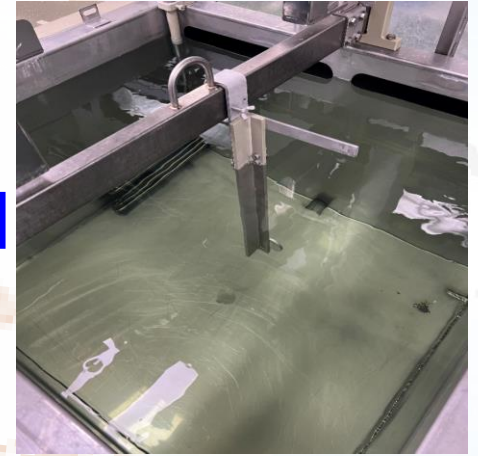
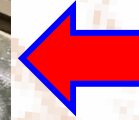
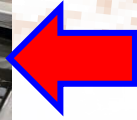
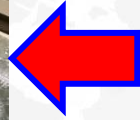
- **Capacity Expansion:** +2 New Plating Tanks (for increased output)
- **Precision Clearance:** Crane side view for accurate calculation of automatic arm to existing pipeline clear distance.
- **Automated Workflow:** Optimal process integration with new equipment.





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Automatic Pretreatment Display



Manual / Automated Comparison

Manual Mode

- Unlocked fixturing (Wobbling)
- Pretreatment time: 45 mins
- Manual control | Accuracy: 80%



Automated Mode

- Threaded fixturing (Stable)
- Pretreatment time: **35 mins**
- PLC control | Accuracy: **>99%**



Manual Mode

AMAT PN	Manual Pretreatment (min)	Batch Size	ENP Time (hr)	ENP Thickness(μm)
0023-00053	45	4	5	28 +/-2
0042-38612	45	2	5	28 +/-2
0022-35279	45	3	5	28 +/-2
0022-40582	45	2	5	28 +/-2
0020-37471	45	4	5	28 +/-2
0042-39426	45	1	5	28 +/-2
0023-17638	45	1	5	28 +/-2
0041-95738	45	3	5	28 +/-2
0022-39846	45	4	5	28 +/-2
0022-31506	45	2	5	28 +/-2
0022-19920G01	45	2	5	28 +/-2
0023-61146	45	2	5	28 +/-2
0023-60786	45	2	5	28 +/-2
0023-60784	45	2	5	28 +/-2
0022-55190	45	4	5	28 +/-2
0022-34394	45	4	5	28 +/-2

Automated Mode

AMAT PN	Automated Pretreatment (min)	Batch Size	ENP Time (hr)	ENP Thickness(μm)
0042-21663	35	6	5	28 +/-2
0042-10617	35	2	5	28 +/-2
0042-93868	35	6	5	28 +/-2
0042-69106	35	6	5	28 +/-2
0040-83120	35	9	5	28 +/-2
0042-76542	35	2	5	28 +/-2
0042-05773	35	4	5	28 +/-2
0042-40005	35	6	5	28 +/-2
0040-75332	35	2	5	28 +/-2
0042-43592	35	3	5	28 +/-2
0043-24694	35	6	5	28 +/-2

2026 AMAT ENP Capacity Assessment

AMAT PN	ENP - Amount
0022-55190	248
0042-21663	233
0042-10617	131
0022-35279	101
0042-38612	87
0022-31506	84
0042-76542	70
0022-30507	54
0023-00053	52
0023-13891	30
0020-37471	25
0042-93868	24
0022-40582	23
0041-79520	14
0042-69106	10
0022-39846	8
0042-05773	6
0042-98929	6
0042-40005	5
0022-34394	4
0023-17638	3
0041-95738	2
0042-05772	2
0042-96886	2
0020-21028	1
0040-83120	1
0041-77395	1
0042-39426	1
SUM	1228

Comparison After Improvement

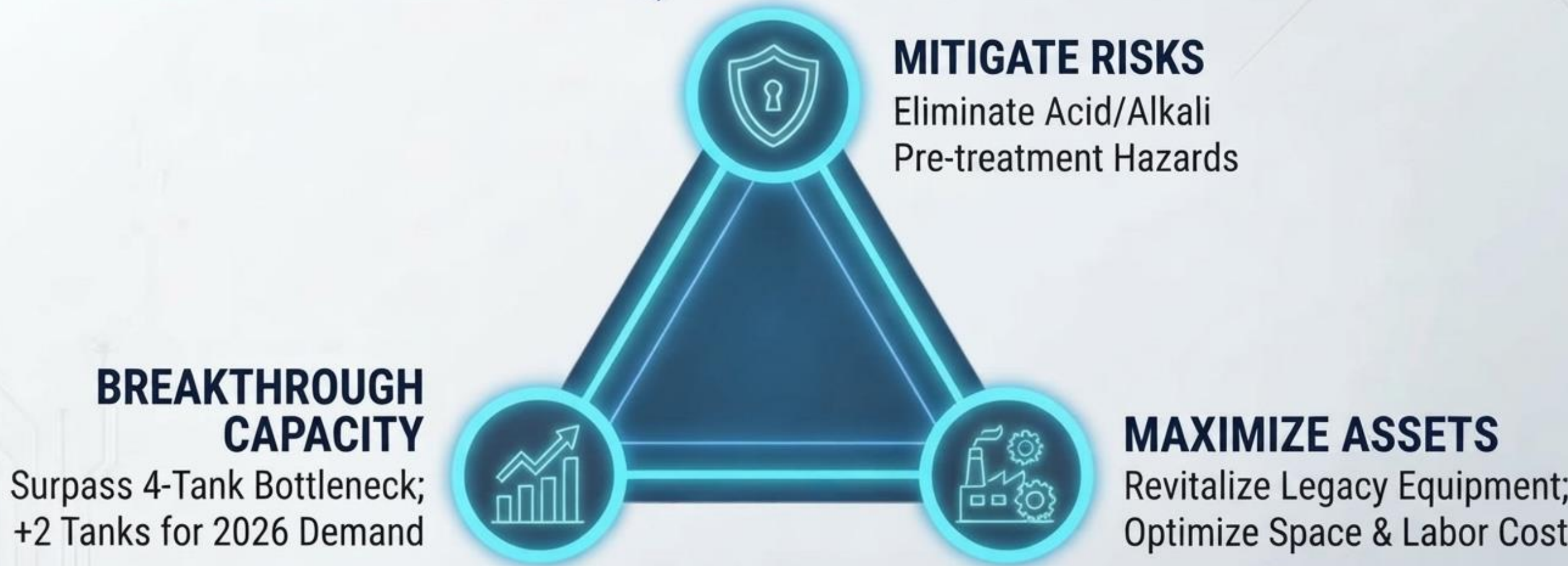
● 2025 Status & Output vs. 2026 Target Capacity Evaluation

Comparison Item	2025 Baseline (Mon–Sat, 1 Day Off)	2026 Target (Before expansion) Headcount Expansion Only	2026 Target (After expansion) Semi-Auto Pretreatment + 2 New Tanks
Plating Tanks	2	2	4
Daily Working Hours (Incl. Masking, Fixturing, Bath Prep, 35000 Wash)	10 hrs (incl. overtime)	23 hrs (Offline prep, dual-tank bath prep + weekday overtime)	23 hrs (Offline prep, dual-tank bath prep + Mon–Sat shift overtime)
Shifts	1	3	3
Weekly Hours	60 hrs	115 hrs	147 hrs
Weekly Output	28 PCS	54 PCS	84 PCS
Monthly Avg. Output	112 PCS	216 PCS	336 PCS
Annual Max. Capacity	1,344 PCS	2,592 PCS	4,032 PCS
Annual Actual Output	1,228 PCS	2,070 PCS (FCST)	3,628 PCS (90%) / 3,225 PCS (80%)
Utilization Rate	91.30%	80%	80~90%

Key Notes:

- 2026 Before expansion target: Max capacity of **2,592** PCS. At 80% utilization, estimated output is **2,070** PCS/year (min).
- 2026 Post-expansion (Auto Pretreatment + 2 Tanks): Max capacity of **4,032** PCS. At 80% utilization, estimated output is **3,225** PCS/year (min).

CONCLUSION: STRATEGIC INVESTMENTS FOR INTELLIGENT & HIGH-QUALITY GROWTH



**A CRITICAL UPGRADE FOR INTELLIGENT, SAFE,
& HIGH-QUALITY ELECTROLESS NICKEL PLATING**